Yaroslav Zhumagulov

Condensed Matter Theory | Quantum Materials | First-Principles-to-Experiment Modeling
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Profile

Condensed matter theorist working on quantum materials. I build low-energy models from first-principles calculations and use them to explain what experiments measure. Most of this work is on graphene heterostructures, where proximity-induced spin-orbit and exchange coupling reshape correlated and superconducting phases, and I write the ab initio code that constructs the models. Earlier projects covered iron-based superconductors, BaBiO3, magnetism in CrI3 monolayers, and many-body excitons and trions in 2D semiconductors. Google Scholar: 433 citations, h-index 11.

Employment

Postdoctoral Researcher / Junior Researcher Oct 2024 – Present
EPFL, Institute of Physics, Chair of Computational Condensed Matter Physics, Lausanne, Switzerland
Supervisor: Prof. Oleg Yazyev

- Developed an ab initio package that builds symmetry-adapted low-energy models of proximity and defect scattering directly from DFT, automating a step usually done by hand.
- Use these models to predict how proximity spin-orbit and exchange coupling change the correlated and superconducting phases of van der Waals materials, working closely with experimental groups.
- Compute the observables that tie the models to experiment: band structures, Wannier Hamiltonians, Berry curvature and topological responses, gaps, and spin-valley splittings.

Visiting Student 2019 – 2020
University of Regensburg, Institute I - Theoretical Physics, Regensburg, Germany

- Developed the research direction on proximity-induced spin interactions, spin-valley physics, and correlated phenomena in two-dimensional quantum materials.

Researcher 2018 – 2019
ITMO University, St. Petersburg, Russia

- Conducted theoretical and numerical research on optical excitations, superconductivity, and strongly correlated materials.

Relevant Technical Evidence

- **Scientific software development:** Author of an ab initio package for automated, symmetry-adapted construction of low-energy effective models of proximity and defect scattering from first-principles calculations, integrating group-theory analysis with DFT/Wannierization workflows.
- **First-principles to models:** Extensive experience with Quantum ESPRESSO, GPAW, Wannier90, and WannierBerri workflows for DFT, Wannierization, band-structure analysis, Berry curvature, and topological-response calculations.
- **Superconductivity and quantum magnetism:** Modeled iron-based superconductors, BaBiO3, magnetic CrI3 monolayers, real-space pairing, Hubbard physics, and superconducting diode phenomenology.

- **Code and HPC:** Python and Fortran for numerical modeling; CuPy/CUDA, MPI, OpenMP, HDF5, Linux/HPC workflows, and batch automation for large numerical simulations and data-heavy calculations.
- **Data for materials discovery:** Produce benchmark datasets from first-principles and model calculations (Hamiltonian parameters, symmetry labels, band structures, response functions) for data-driven materials modeling.
- **Theory-experiment collaboration:** Translate microscopic models into experimentally testable predictions, including gaps, spin-valley observables, magnetic responses, optical spectra, and correlated phase boundaries.

Education

PhD in Theoretical Physics Oct 2020 – Nov 2023
University of Regensburg, Institute I - Theoretical Physics, Germany
magna cum laude
Thesis: Theoretical investigation of the interplay of proximity-induced spin interactions and correlated phenomena in multilayer graphene systems. Supervisor: Prof. Jaroslav Fabian.

MSc in Condensed Matter Physics and Photonics Sep 2016 – Jun 2018
National Research Nuclear University MEPhI, Department of Solid State Physics and Nanosystems, Russia
GPA: 4.9/5.0
Thesis: Investigation of magnetic, nematic and superconducting properties of high-temperature iron-based superconductors using the variational cluster approximation method. Supervisor: Dr. Andrey Krasavin.

BSc in Condensed Matter Physics and Photonics Sep 2012 – Jun 2016
National Research Nuclear University MEPhI, Department of Solid State Physics and Nanosystems, Russia
GPA: 4.3/5.0
Thesis: Recovery of the density of states of strongly correlated systems using numerical methods. Supervisor: Dr. Andrey Krasavin.

Selected Publications

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1. **Y. Zhumagulov**, D. Kochan, and J. Fabian, "Emergent correlated phases in rhombohedral trilayer graphene induced by proximity spin-orbit and exchange coupling," *Physical Review Letters* 132, 186401 (2024).

2. **Y. Zhumagulov**, D. Kochan, and J. Fabian, "Swapping exchange and spin-orbit induced correlated phases in proximitized Bernal bilayer graphene," *Physical Review B* 110, 045427 (2024).

3. A. M. Seiler, **Y. Zhumagulov**, K. Zollner, C. Yoon, D. Urbaniak, F. R. Geisenhof, K. Watanabe, T. Taniguchi, J. Fabian, F. Zhang, and R. Weitz, "Layer-selective spin-orbit coupling and strong correlation in bilayer graphene," *2D Materials* 12(3), 035009 (2025).

4. **Y. Zhumagulov**, S. Chiavazzo, I. A. Shelykh, and O. Kyriienko, "Robust polaritons in magnetic monolayers of CrI3," *Physical Review B* 108, L161402 (2023).

5. A. Kudlis, M. Kazemi, **Y. Zhumagulov**, H. Schrautzer, A. I. Chernov, P. F. Bessarab, I. V. Iorsh, and I. A. Shelykh, "All-optical magnetization control in CrI3 monolayers: A microscopic theory," *Physical Review B* 108, 094421 (2023).

6. A. P. Menushenkov, A. Ivanov, V. Neverov, A. Lukyanov, A. Krasavin, A. Yastrebtsev, Y. Kovalev, **Y. Zhumagulov**, A. Kuznetsov, V. Popov, and others, "Direct evidence of real-space pairing in BaBiO3," *Physical Review Research* 6, 023307 (2024).

7. A. E. Lukyanov, I. A. Kovalev, V. D. Neverov, **Y. V. Zhumagulov**, A. V. Krasavin, and D. Kochan, “Strength of the Hubbard potential and its modification by breathing distortion in BaBiO3,” *Physical Review B* 105, 045131 (2022).

8. A. E. Lukyanov, V. D. Neverov, **Y. V. Zhumagulov**, A. P. Menushenkov, A. V. Krasavin, and A. Vagov, “Laser-induced ultrafast insulator-metal transition in BaBiO3,” *Physical Review Research* 2, 043207 (2020).

9. D. Kochan, A. Costa, **I. Zhumagulov**, and I. Zutic, “Phenomenological theory of the supercurrent diode effect: The Lifshitz invariant,” arXiv:2303.11975 (2023).

10. **Y. V. Zhumagulov**, V. A. Kashurnikov, A. V. Krasavin, A. Lukyanov, and V. D. Neverov, “Phase diagram of the two-orbital model for iron-based HTSC: Variational cluster approximation,” *JETP Letters* 109, 45 (2019).

Grants, Honors, and Awards

- Best oral presentation, 5th International Youth Scientific Conference “Physics. Technologies. Innovations PTI-2018” (2018).
- Increased scholarship for activities in science, National Research Nuclear University MEPhI (2015 – 2018).
- First prize, Competition of Youth Scientific Works, P. N. Lebedev Physical Institute of the Russian Academy of Sciences and Moscow Engineering Physics Institute (2016).